

## **Basic Buck Converter — Step-Down DC–DC Conversion**

## Objective

This note examines the open-loop behavior of a DC–DC buck converter using PSIM simulation. The focus is on:

- The relationship between duty cycle and average output voltage.
- The influence of inductance and capacitance on ripple and startup transient response.

The goal is to observe fundamental converter behavior before introducing closed-loop control.

## System Model

Input voltage: 50 V DC

Load: 5  $\Omega$  resistive

Control method: Open-loop PWM

Duty ratios evaluated:  $D = 0.3, 0.5, 0.7$

Simulations were performed in PSIM to compare voltage and inductor-current waveforms under identical hardware conditions.

## Basic Principle

Operating Modes

### When the switch is ON:

The input voltage is applied to the inductor, and energy is stored in its magnetic field.

$$V_L = V_{in} - V_{out}$$

### When the switch is OFF:

The inductor releases stored energy through the diode into the load, maintaining continuous current flow.

$$V_L = -V_{out}$$

The average output voltage is given by:

$$V_{out} = D * V_{in}$$

where  $D$  is the duty cycle.

**Example:** For  $V_{in} = 50V$  and  $D = 0.5$ ,  $V_{out} \approx 25V$ .

## Circuit Parameters

| PARAMETER | SYMBOL | VALUE | DESCRIPTION |
|-----------|--------|-------|-------------|
|-----------|--------|-------|-------------|

|                     |              |               |                            |
|---------------------|--------------|---------------|----------------------------|
| INPUT VOLTAGE       | ( $V_{in}$ ) | 50 V          | Supply voltage             |
| LOAD RESISTANCE     | ( $R$ )      | 5 $\Omega$    | Represents load            |
| INDUCTOR            | ( $L$ )      | 1 mH          | Controls current ripple    |
| CAPACITOR           | ( $C$ )      | 100 $\mu$ F   | Smooths voltage ripple     |
| SWITCHING FREQUENCY | ( $f_s$ )    | 5 kHz         | Determines ripple and loss |
| DUTY CYCLE          | ( $D$ )      | 0.3, 0.5, 0.7 | Target ratio               |

**These values were selected to maintain continuous conduction while allowing visible ripple and transient effects.**

## Simulation Observations

### Steady-State Behavior

- The average output voltage scaled proportionally with duty cycle.
- $D = 0.3$  produced the lowest output level.
- $D = 0.5$  produced approximately 25 V.
- $D = 0.7$  produced the highest output level among the tested cases.

The proportional relationship between  $V_{out}$  and  $D$  was observed in the waveform comparison (Figures 2 and 3).

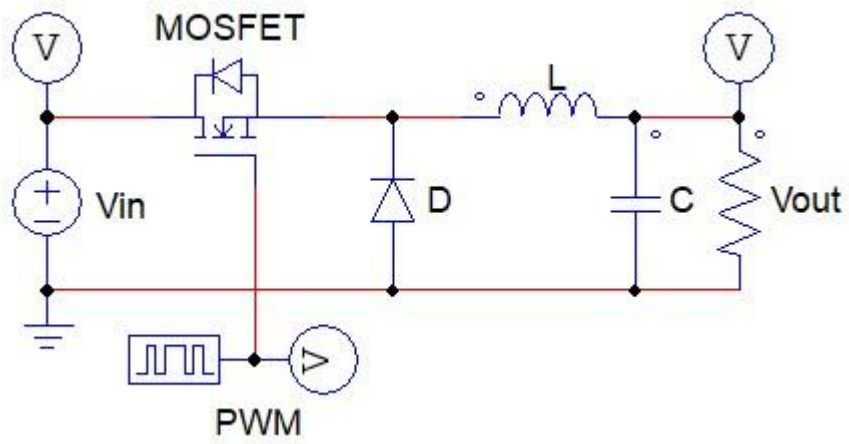


Figure 1

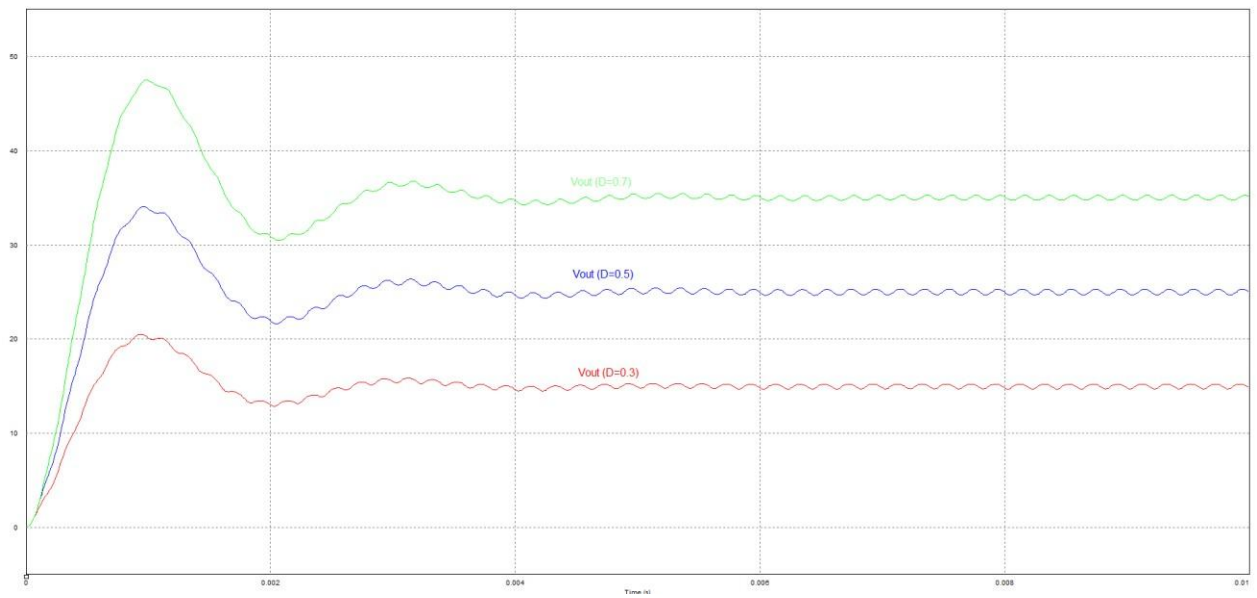
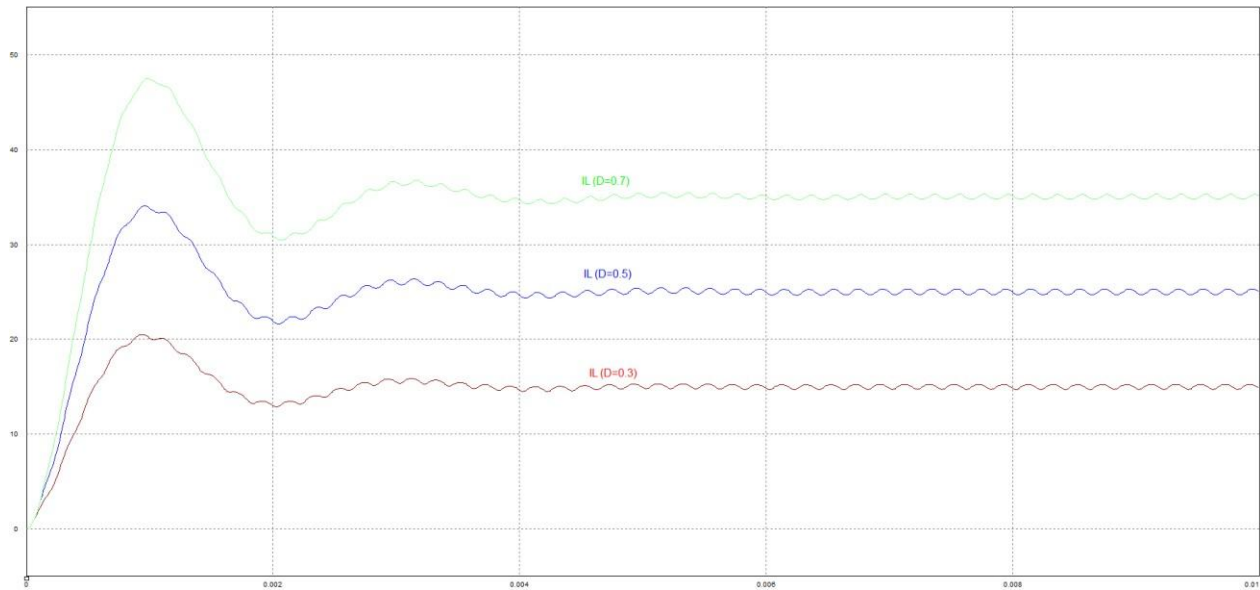


Figure 2 Comparison of output voltage waveforms at duty cycles  $D = 0.3, 0.5$ , and  $0.7$ .



*Figure 3 Comparison of inductor's current waveforms at duty cycles  $D = 0.3$ ,  $0.5$ , and  $0.7$ .*

## Startup Behavior

With  $L = 1 \text{ mH}$  and  $C = 100 \text{ }\mu\text{F}$ :

- A noticeable overshoot occurred before settling.
- Inductor current exhibited a corresponding transient peak.

When  $L$  was increased to  $5 \text{ mH}$  and  $C$  reduced to  $30\text{--}50 \text{ }\mu\text{F}$ :

- The voltage rise became smoother.
- Overshoot was significantly reduced.

This indicates that transient shape is strongly dependent on energy storage distribution between  $L$  and  $C$ .

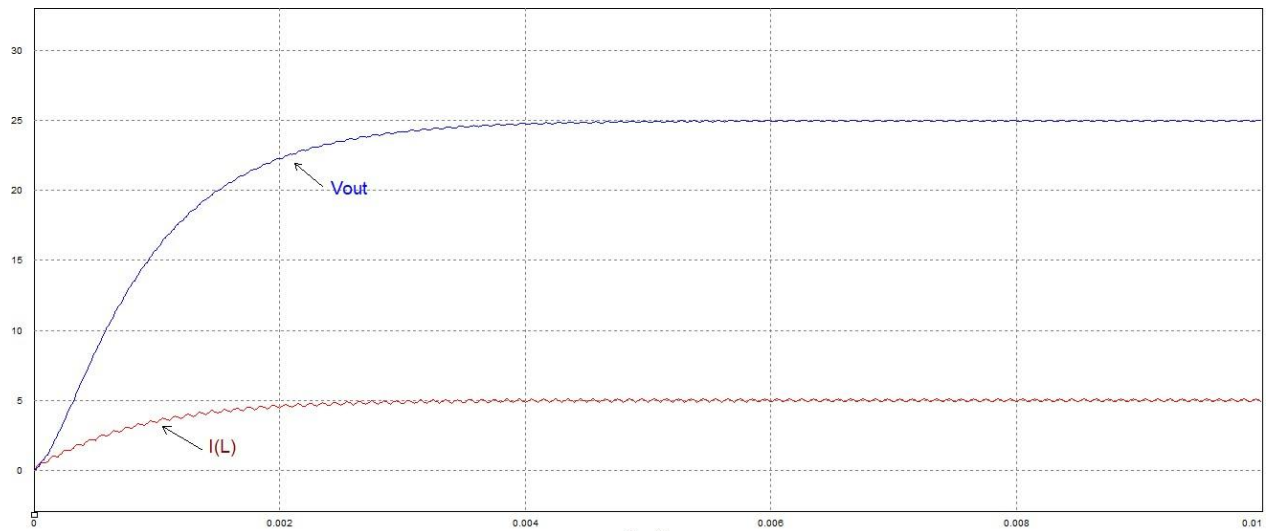


Figure 4

## Parameter Sensitivity

### Duty Cycle

- Directly controls the average output voltage in open-loop operation.
- Higher  $D$  increases both output voltage and inductor current stress.

### Inductance

- Larger  $L$  reduces current ripple.
- Larger  $L$  slows dynamic response.

### Capacitance

- Larger  $C$  reduces voltage ripple.
- Excessive  $C$  can increase startup overshoot if inductor energy transfer is not damped.

## Summary

- In open-loop CCM operation,  $V_{out}$  follows  $V_{out} = D \cdot V_{in}$ .
- Increasing duty cycle proportionally increases average output voltage.
- Startup overshoot is primarily determined by the  $L$ - $C$  network under open-loop operation.
- Larger inductance improves damping but slows response.
- Converter transient behavior is sensitive to passive component selection even without feedback control.

This analysis establishes the open-loop behavior baseline required before implementing voltage or current feedback control.